

EDC — Chapter 1: The Bipolar Junction Transistor (BJT)

Electronic Devices and Circuit (ENEX 151 / EX 151) · Tribhuvan University, IOE · BE (BEI, BCT) I/II

Sources synthesized: Sedra-Smith (Microelectronic Circuits), Boylestad-Nashelsky (Electronic Devices and Circuit Theory), Floyd (Electronic Devices), Millman-Halkias (Integrated Electronics), Bell (Electronic Devices and Circuits) + your 4 PYQ papers (2081 Ashwin, 2082 Baishakh, 2082 Bhadra, 2083 Baishakh).

Chapter Overview — Read This First

What this chapter is about: The BJT is a three-terminal semiconductor device in which a small base current (or base-emitter voltage) controls a much larger collector current. Everything in this chapter builds one story:

Physics of the device → its I-V characteristics → setting a DC operating point (biasing) → replacing the transistor with a small-signal model → analyzing the three amplifier configurations → using the BJT as a switch → the complete large-signal (Ebers-Moll) model.

DATA PYQ Frequency Analysis (from your 4 papers)

BJT contributed **exactly 13 marks out of 60 in every one of the four papers** — it is the single most predictable chapter on this paper.

Topic	2081 Ashwin	2082 Baishakh	2082 Bhadra	2083 Baishakh	Verdict
β -independent bias design (voltage divider)	✓ [4] CE	✓ [4]	✓ [1+5]	✓ [1+5] CC	HOT 4/4 years — GUARANTEED
Small-signal analysis of one configuration (Z_i , Z_o , A_v)	✓ CC [1+4]	✓ CE numerical [5]	✓ CE [2+5]	✓ CB [2+5]	HOT 4/4 years — GUARANTEED
Role of coupling / bypass capacitors	–	–	✓ coupling [2]	✓ bypass [2]	HOT 2/4 — attached to the amplifier question
BJT as a switch	–	✓ [4]	–	–	● 1/4
Derive transconductance g_m	✓ [4]	–	–	–	● 1/4
Why BJT is bipolar / FET unipolar	–	–	–	✓ [1]	● 1/4
Load line, DC analysis, characteristics	(embedded inside design/numerical questions)				● foundational
Ebers-Moll model	never asked				● low, but on syllabus

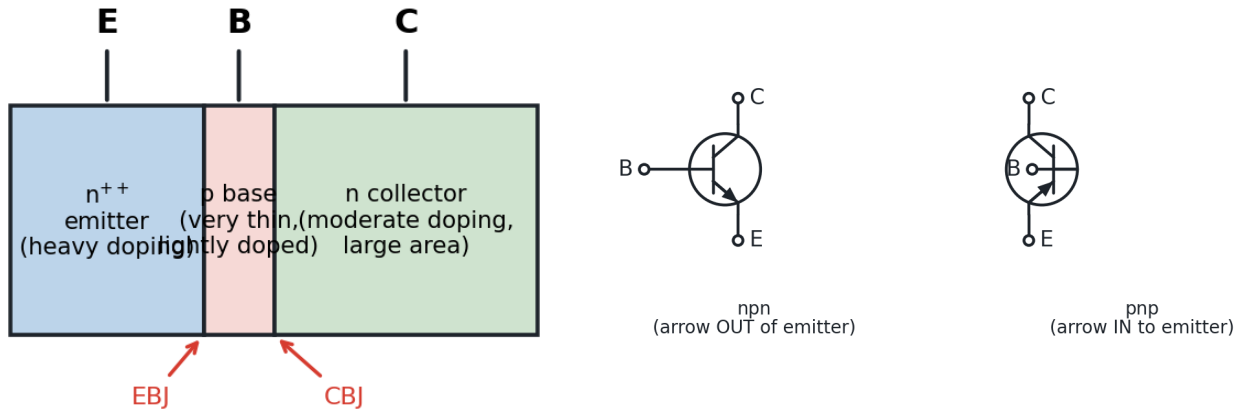
Strategic conclusion: 1. **HOT** You **must** be able to *design* a voltage-divider (β -independent) bias circuit for **any** configuration (CE, CC) with any given V_{CC} , I_C/I_E , β — this has appeared **every single year**. 2. **HOT** You **must** be able to draw the small-signal equivalent and *derive* Z_i , Z_o , A_v for **all three configurations** — the paper rotates the configuration each year (CE → CC → CE → CB). After CB in

2083, **CE and CC are both live candidates**. 3. ● Prepare the 4-mark "side" theory: g_m derivation, BJT as switch, capacitor roles, bipolar-vs-unipolar.

1.1 Review of Operation of the npn Transistor in the Active Mode

A. Intuition — what the device is

A BJT is two pn junctions built back-to-back sharing a **very thin, lightly doped base**:



nnp structure — heavily doped emitter, very thin lightly doped base, large collector — and the npn/pnp circuit symbols (arrow always on the emitter, pointing along conventional emitter current).

Three structural facts explain everything:

1. **Emitter is heavily doped (n^{++})** → it can inject a huge number of electrons.
2. **Base is extremely thin and lightly doped** → electrons injected into it almost all survive the trip across; very few recombine.
3. **Collector is large and moderately doped** → it collects the electrons and handles power dissipation.

The transistor is NOT two independent diodes. Two discrete diodes wired back-to-back give no transistor action, because the thin shared base is what lets carriers injected by one junction be collected by the other.

B. The four modes of operation

The mode is set by how the two junctions are biased:

Mode	EBJ (B-E)	CBJ (B-C)	Use
Active	Forward	Reverse	Amplifier ← this section
Saturation	Forward	Forward	Switch ON
Cutoff	Reverse	Reverse	Switch OFF
Reverse-active	Reverse	Forward	Rarely used (poor β)

Active-mode bias conditions (nnp): $V_{BE} \approx 0.7 \text{ V}$ (forward), $V_{CB} \geq 0$ — equivalently $V_{CE} > V_{CE(sat)} \approx 0.2\text{--}0.3 \text{ V}$. Memory aid for npn active mode: $V_C > V_B > V_E$.

C. Physical operation in the active mode (formal explanation)

With the EBJ forward-biased and CBJ reverse-biased:

1. **Injection:** The forward-biased EBJ lowers the junction barrier; the heavily doped emitter injects a large flood of **electrons into the base** (and the base injects a few holes into the emitter — this hole current is small because the base is lightly doped).

2. **Diffusion across the base:** Injected electrons become minority carriers in the p-type base and **diffuse** toward the collector. Because the base is thin, the diffusion transit is short.
3. **Little recombination:** Only a small fraction of electrons recombine with holes in the base. The holes lost to recombination (plus the holes injected into the emitter) must be resupplied by the base terminal → this constitutes the **small base current I_B** .
4. **Collection:** Electrons reaching the reverse-biased CBJ are swept into the collector by the junction field → **large collector current I_C** , almost equal to the injected electron current.

Key insight: I_C is controlled by V_{BE} (which sets injection), and is nearly independent of V_{CB}/V_{CE} — the collector merely *collects*. This is why the active-mode BJT behaves as a **voltage-controlled current source**.

Why is the BJT called "bipolar"? HOT PYQ 2083 Q1(a) [1 mark]

Current conduction in a BJT involves **both types of charge carriers — electrons AND holes** (e.g., in an npn device, electrons carry the main emitter→collector current while holes constitute the base current and recombination). Hence **bi-polar**. In an FET, the channel current is carried by **only one carrier type** (electrons in n-channel, holes in p-channel), so the FET is **uni-polar**. A complete 1-mark answer states both halves.

D. Current relations and equations

Terminal current balance (KCL on the device):

- $I_E = I_C + I_B$ — always true, all modes, both npn and pnp.

Common-base current gain α :

- $I_C = \alpha \cdot I_E$, where α = fraction of emitter current collected. Typically $\alpha = 0.98 - 0.998$ ($\alpha < 1$, very close to 1).

Common-emitter current gain β (also h_{FE}):

- $I_C = \beta \cdot I_B$, where $\beta = 50 - 300$ typically (take $\beta \approx 100$ unless told otherwise).

$\alpha \leftrightarrow \beta$ conversion (derive once, remember forever):

Start from $I_E = I_C + I_B$ and $I_C = \alpha I_E$: $I_C = \alpha(I_C + I_B) \rightarrow I_C(1 - \alpha) = \alpha I_B \rightarrow I_C = [\alpha/(1-\alpha)] \cdot I_B$

- $\beta = \alpha/(1 - \alpha)$ and inverting: $\alpha = \beta/(\beta + 1)$
- Also: $I_E = (\beta + 1) \cdot I_B$

Exponential collector current law (Sedra-Smith — the foundation of g_m):

- $I_C = I_S \cdot e^{V_{BE}/V_T}$

Symbol	Meaning	Typical value
I_S	Saturation (scale) current, $\propto$ junction area	$10^{-12} - 10^{-15}$ A
V_T	Thermal voltage = kT/q	$\approx 25\text{--}26$ mV at room temperature
V_{BE}	Base-emitter voltage	≈ 0.7 V in conduction

Conditions: active mode only; ignores Early effect (add factor $(1 + V_{CE}/V_A)$ to include it).

E. Exam interpretation

Examiners test: (i) the bipolar/unipolar distinction, (ii) junction bias conditions per mode, (iii) fluency with α , β , $I_E = I_C + I_B$ in numericals, (iv) the physical story (injection → diffusion → collection) as a descriptive answer.

F. Common mistakes

- Writing $\beta = \alpha/(1 + \alpha)$. **Wrong** — denominator is $(1 - \alpha)$.

- Saying I_C flows "because CBJ is forward biased." No — CBJ is **reverse** biased in active mode; it *collects*.
- Treating the BJT as two diodes when explaining transistor action.
- Forgetting that for **pn**p everything reverses: $V_{EB} \approx 0.7\text{ V}$, currents flow opposite, memory aid $V_E > V_B > V_C$ for active mode.

KEY Section Summary

- **Understand:** injection-diffusion-collection story; why thin base $\Rightarrow \beta$ large; bipolar vs unipolar.
- **Memorize:** mode table; $I_E = I_C + I_B$; $I_C = \beta I_B = \alpha I_E$; $\beta = \alpha/(1-\alpha)$; $I_C = I_S \cdot e^{V_{BE}/V_T}$.
- **Draw:** npn structure with junctions; npn/pnp symbols with current directions.

PYQ Check

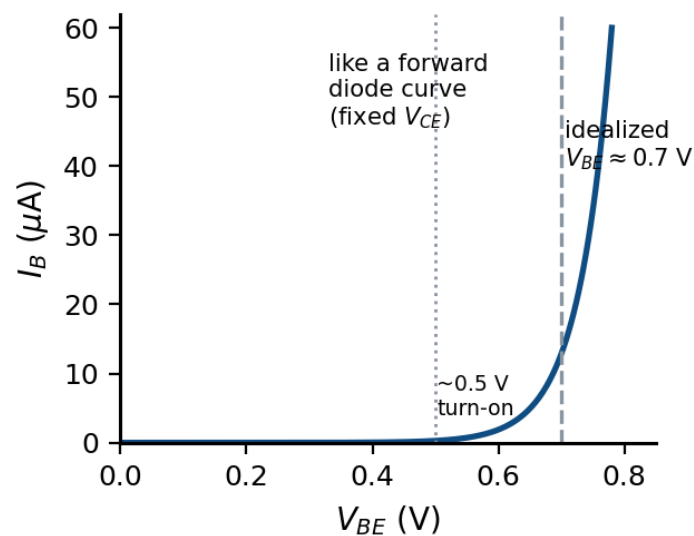
- 2083 Baishakh Q1(a): *Why BJT is bipolar and FET is unipolar?* [1] — answered above.

Exam-ready when: you can explain active-mode operation in 5-6 sentences with a structure sketch, and interconvert α , β , I_B , I_C , I_E instantly.

1.2 Review of Graphical Representation of Transistor Characteristics

A. Input (base) characteristics — CE configuration

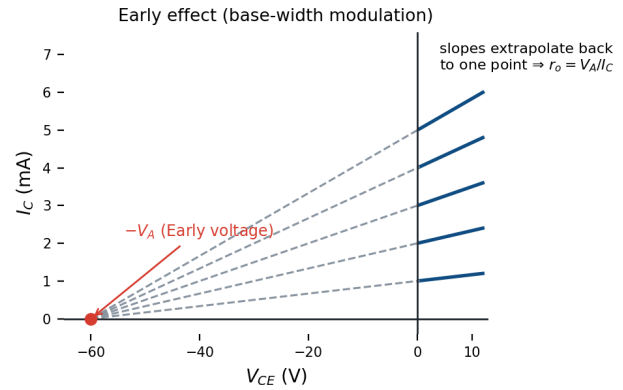
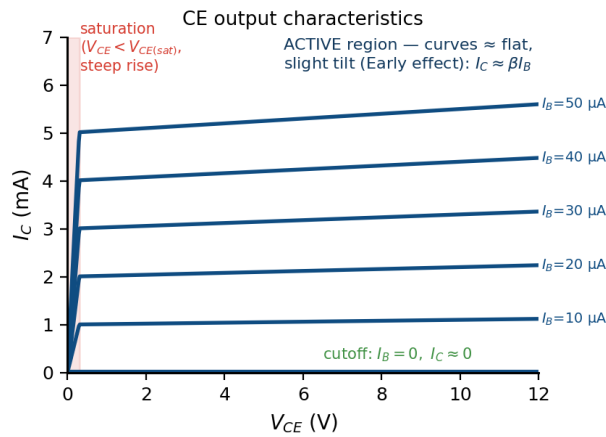
Plot of I_B vs V_{BE} (for fixed V_{CE}): looks exactly like a forward diode curve. Negligible I_B until $V_{BE} \approx 0.5\text{ V}$, then rapid exponential rise; in analysis we idealize $V_{BE} \approx 0.7\text{ V}$ whenever the transistor conducts.



Input (base) characteristic I_B vs V_{BE} for fixed V_{CE} — a forward-diode curve, idealized to $V_{BE} \approx 0.7\text{ V}$ in analysis.

B. Output (collector) characteristics — CE configuration •

Family of I_C vs V_{CE} curves, one curve per value of I_B :



CE output characteristics: one I_C - V_{CE} curve per I_B , with cutoff, saturation and active regions (left); active-region slopes extrapolate back to $-V_A$, giving $r_o = V_A / I_C$ (right).

Exam diagram — practice drawing this. Label the three regions:

Region	Condition	Behaviour
Cutoff	$I_B = 0$ (EBJ not forward)	$I_C \approx 0$; transistor OFF
Saturation	$V_{CE} < V_{CE(sat)} \approx 0.2\text{--}0.3\text{ V}$	Both junctions forward; $I_C < \beta I_B$; curves rise steeply
Active	$V_{CE} > V_{CE(sat)}$, $I_B > 0$	$I_C \approx \beta I_B$, nearly independent of $V_{CE} \rightarrow$ amplifier region

C. Early effect (base-width modulation)

In the active region the curves are not perfectly flat: increasing V_{CE} widens the CBJ depletion region, **narrowing the effective base**, which slightly increases I_C . Extrapolated backwards, all active-region curves meet at $V_{CE} = -V_A$, the **Early voltage** ($V_A \approx 50\text{--}150\text{ V}$).

Including it: $I_C = I_S \cdot e^{V_{BE}/V_T} \cdot (1 + V_{CE}/V_A)$

Consequence — the transistor has a finite **output resistance**:

- $r_o = V_A / I_C$ (typically tens of k Ω ; appears later in the small-signal model, and as the given " $r_o = 40\text{ k}\Omega$ " in PYQ 2082 Baishakh Q2)

D. Common mistakes

- Confusing BJT "saturation region" (V_{CE} small, switch ON) with MOSFET "saturation" (the *amplifying* region). Opposite meanings!
- Drawing active-region curves perfectly flat and then being unable to explain r_o .

KEY Section Summary

- **Draw:** output characteristic family with 3 labelled regions and Early-voltage extrapolation.
- **Memorize:** $r_o = V_A / I_C$; $V_{CE(sat)} \approx 0.2\text{ V}$; region conditions.
- No standalone PYQ, but this diagram is **required inside** load-line, switch, and r_o -based answers.

1.3 Analysis of Transistor Circuits at DC

A. The universal 5-step procedure ●

Given any DC BJT circuit (capacitors $\rightarrow$ open at DC):

1. **Assume active mode** $\rightarrow$ set $V_{BE} = 0.7\text{ V}$ and use $I_C = \beta I_B$, $I_E = (\beta + 1)I_B$.
2. Write **KVL around the base-emitter loop** $\rightarrow$ solve for I_B (or I_E).
3. Compute I_C , then **KVL around the collector loop** $\rightarrow V_{CE}$.

4. **Verify the assumption:** active requires $V_{CE} > 0.3 \text{ V}$ (npn). If V_{CE} comes out $< 0.3 \text{ V}$ (or negative) → transistor is actually **saturated**: redo with $V_{CE} = 0.2 \text{ V}$, $I_C \neq \beta I_B$.
5. If EBJ not forward-biased at all → **cutoff**: all currents = 0.

B. Worked example (fixed bias with emitter resistor — the exact DC front-end of PYQ 2082 Baishakh Q2)

Given: $V_{CC} = 20 \text{ V}$, $R_B = 470 \text{ k}\Omega$ (base to V_{CC}), $R_C = 2.2 \text{ k}\Omega$, $R_E = 0.56 \text{ k}\Omega$, $\beta = 120$.

Base loop KVL: $V_{CC} = I_B \cdot R_B + V_{BE} + I_E \cdot R_E$, with $I_E = (\beta + 1)I_B$:

$$I_B = (V_{CC} - V_{BE}) / [R_B + (\beta + 1)R_E] = (20 - 0.7) / (470\text{k} + 121 \times 0.56\text{k}) = 19.3 / 537.76\text{k} = \mathbf{35.9 \mu A}$$

$$I_C = \beta I_B = 120 \times 35.9 \mu A = \mathbf{4.31 \text{ mA}}; I_E = (\beta + 1)I_B \approx \mathbf{4.34 \text{ mA}}$$

Collector loop: $V_{CE} = V_{CC} - I_C \cdot R_C - I_E \cdot R_E \approx 20 - 4.31\text{mA} \times 2.2\text{k} - 4.34\text{mA} \times 0.56\text{k} = 20 - 9.48 - 2.43 = \mathbf{8.09 \text{ V}} \rightarrow > 0.3 \text{ V} \checkmark$ active confirmed.

This DC result ($I_E \approx 4.34 \text{ mA} \rightarrow r_e \approx 6 \Omega$) is **step 1 of the 5-mark small-signal PYQ** solved fully in §1.9.

C. Common mistakes

- Forgetting the $(\beta + 1)R_E$ reflection term in the base loop — the #1 DC error.
- Not verifying the active assumption.
- Using I_C where I_E belongs in the emitter-resistor drop (acceptable approximation only if stated: $I_E \approx I_C$).

KEY Section Summary

- **Must be able to calculate:** I_B , I_C , I_E , V_{CE} for fixed-bias, emitter-bias, and voltage-divider circuits (divider case in §1.6).
- DC analysis is a **prerequisite embedded in every design and amplifier PYQ** — it silently carries 2–3 marks of every 5-mark question.

1.4 Graphical DC Load Line Analysis

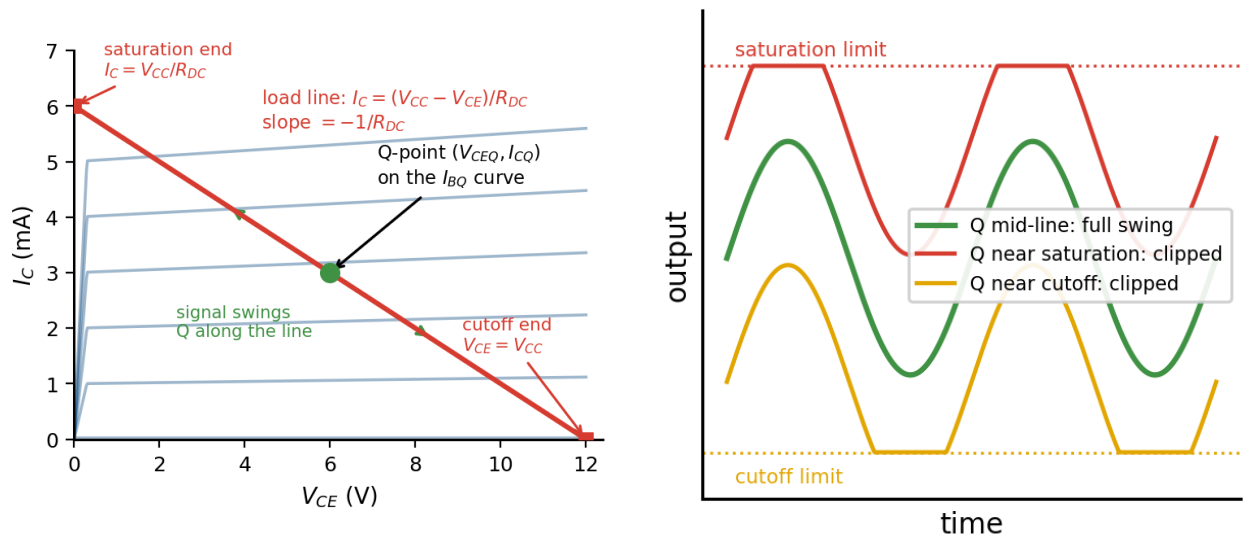
A. Concept

The transistor obeys its output characteristics; the external collector circuit obeys KVL. The operating point must satisfy **both simultaneously** → intersect them graphically.

Collector-loop KVL (CE stage, total DC resistance $R_{DC} = R_C + R_E$):

$$V_{CE} = V_{CC} - I_C \cdot R_{DC} \rightarrow \text{rearranged: } I_C = (V_{CC} - V_{CE}) / R_{DC} \text{ — a straight line on the output characteristics.}$$

Endpoints: - V_{CE} -axis intercept ($I_C = 0$): $V_{CE} = V_{CC} \rightarrow$ cutoff end. - I_C -axis intercept ($V_{CE} = 0$): $I_{C(sat)} = V_{CC} / R_{DC} \rightarrow$ saturation end. - Slope = $-1/R_{DC}$.



DC load line $I_C = (V_{CC} - V_{CE})/R_{DC}$ drawn over the characteristics; Q is its intersection with the I_{BQ} curve (left). Mid-line Q gives maximum symmetric swing — Q near either end clips (right).

Exam diagram — practice drawing this (load line drawn on top of the §1.2 family).

B. Using the load line

- **Q-point (quiescent point):** the DC operating point (I_{CQ} , V_{CEQ}) with no signal applied. Set by the bias circuit (I_{BQ}) + load line.
- **Signal swing:** an AC input wiggles $I_B \rightarrow$ the operating point slides along the load line. Q centred near the middle ($V_{CEQ} \approx V_{CC}/2$) gives **maximum symmetric swing** — this is exactly why bias-design rules in §1.6 choose $V_{CE} = V_{CC}/2$.
- Q too near saturation $\rightarrow$ negative peaks clip; too near cutoff $\rightarrow$ positive peaks clip.
- Changing R_C rotates the line (slope); changing V_{CC} shifts it; changing I_B moves Q along it.

KEY Section Summary

- **Memorize:** endpoints (V_{CC} and V_{CC}/R_{DC}); slope $-1/R_{DC}$; "mid-point bias $\Rightarrow$ max swing".
- **Draw:** load line + Q + clipping argument.
- No standalone PYQ, but it is the **justification you quote inside every bias-design answer** (" $V_{CE} = V_{CC}/2$ for maximum symmetrical swing" earns method marks).

1.5 Transistor as an Amplifier (r_π , r_e , g_m)

A. Intuition

Bias the BJT in active mode at Q, then superimpose a tiny wiggle v_{be} on top of V_{BE} . Because I_C depends **exponentially** on V_{BE} , a millivolt-scale wiggle produces a milliamp-scale wiggle in I_C . Push that current wiggle through a collector resistor and you get a **large voltage wiggle** $\rightarrow$ amplification. The three parameters g_m , r_π , r_e are just the linearized "exchange rates" between the small signals.

B. HOT Derivation of transconductance g_m — PYQ 2081 Ashwin Q3 [4 marks]

Objective: find $g_m \equiv i_c / v_{be}$ (collector-current response to base-emitter voltage), at bias point I_C .

Step 1 — start from the fundamental law (active mode): $i_C = I_S \cdot e^{v_{BE}/V_T}$

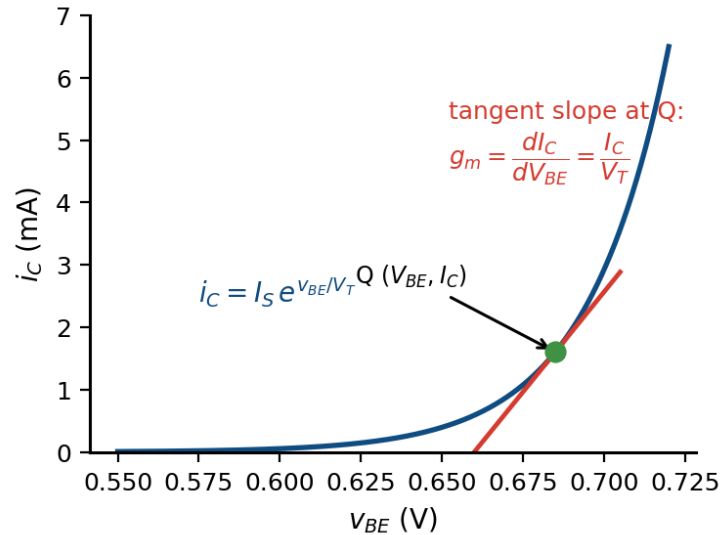
Step 2 — split total quantities into DC + signal: $v_{BE} = V_{BE} + v_{be}$: $i_C = I_S \cdot e^{(V_{BE} + v_{be})/V_T} = [I_S \cdot e^{V_{BE}/V_T}] \cdot e^{v_{be}/V_T} = I_C \cdot e^{v_{be}/V_T}$

Step 3 — small-signal approximation. For $v_{be} \ll V_T$ (in practice $v_{be} < \sim 10$ mV), expand $e^x \approx 1 + x$: $i_C \approx I_C (1 + v_{be}/V_T) = I_C + (I_C/V_T) \cdot v_{be}$

Step 4 — identify the signal component: $i_c = i_c - I_C = (I_C / V_T) \cdot v_{be}$. Therefore:

$$g_m = i_c / v_{be} = I_C / V_T \quad \checkmark \text{ final result — box this in the exam.}$$

Equivalently $g_m = di_C/dv_{BE}$ evaluated at the Q-point — the **slope of the exponential i_C - v_{BE} curve at Q**.



g_m is the slope of the exponential i_C - v_{BE} law at the Q-point: $g_m = I_C/V_T$.

Numbers: $V_T \approx 25 \text{ mV} \Rightarrow g_m \approx 40 \cdot I_C$ (in mA/V if I_C in mA). E.g., $I_C = 1 \text{ mA} \rightarrow g_m = 40 \text{ mA/V}$.

Exam interpretation: the 4 marks are for (i) the exponential law, (ii) the DC+AC split, (iii) the linearization with its validity condition $v_{be} \ll V_T$, (iv) the boxed result with slope interpretation. Skipping step 3's condition loses a mark.

C. Input resistances r_π and r_e

r_π — resistance seen looking into the base (signal v_{be} , signal current i_b): $i_b = i_c/\beta = (g_m/\beta) \cdot v_{be} \Rightarrow$

$$r_\pi = v_{be} / i_b = \beta / g_m = \beta \cdot V_T / I_C = V_T / I_B$$

r_e — resistance seen looking into the emitter (signal v_{be} across it, but current i_e): $i_e = i_c/\alpha = (g_m/\alpha) \cdot v_{be} \Rightarrow$

$$r_e = v_{be} / i_e = \alpha / g_m \approx 1/g_m = V_T / I_E \approx 25 \text{ mV} / I_E$$

Relations to memorize (used constantly in §1.9):

- $r_\pi = (\beta + 1) \cdot r_e \approx \beta \cdot r_e$
- $g_m \cdot r_\pi = \beta$; $g_m \cdot r_e = \alpha \approx 1$
- $r_e \approx 26 \text{ mV} / I_E (\text{mA}) \Omega \leftarrow$ Boylestad's working formula (26 mV variant; 25 mV also accepted — state which you use)

D. Amplification in one line

With collector resistor R_C : $v_o = -i_c \cdot R_C = -g_m \cdot R_C \cdot v_{be} \Rightarrow \mathbf{A_v = -g_m \cdot R_C}$. Voltage gain is transconductance $\times$ load — the seed of all of §1.9.

FORMULAS Formula Box — §1.5

Formula	What it is	Use / condition
$g_m = I_C/V_T \approx 40 \cdot I_C$	Transconductance	Active mode, Q known; the "gain engine"

Formula	What it is	Use / condition
$r_{\pi} = \beta/g_m = \beta V_T/I_C$	Base input resistance (π -model)	CE input-impedance calcs
$r_e = \alpha/g_m \approx V_T/I_E \approx 26\text{mV}/I_E$	Emitter resistance (T-model)	CB/CC and quick CE calcs
$r_{\pi} = (\beta+1)r_e \approx \beta r_e$	Model bridge	Converting between π and T results
$r_o = V_A/I_C$	Output resistance	Only when V_A or r_o given

Common mistakes

- Using **total** V_{BE} (0.7 V) instead of I_C/V_T when computing g_m .
- Mixing r_{π} and r_e (r_{π} is $\sim\beta$ times larger!).
- Forgetting that all three parameters **depend on the Q-point** — you must do DC analysis first.

PYQ Check

- 2081 Ashwin Q3: *Derive trans-conductance of BJT.* [4] — full derivation in B above.

Exam-ready when: you can reproduce the g_m derivation from a blank page in ~ 6 lines and compute g_m , r_{π} , r_e in seconds from any Q-point.

1.6 Biasing BJT for Discrete-Circuit Design HOT HOT (GUARANTEED QUESTION — 4/4 YEARS)

A. Why bias, and what "good bias" means

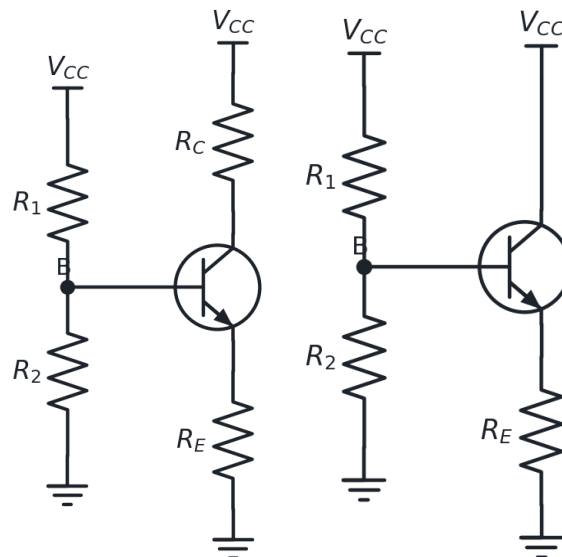
Biasing establishes a stable Q-point (I_{CQ} , V_{CEQ}) so the transistor amplifies linearly. The enemy: **β is unreliable** — it varies unit-to-unit (e.g., 50–150 for the "same" transistor) and with temperature. A good bias circuit makes **I_{CQ} insensitive to β** (and to V_{BE} drift and I_{CBO}).

B. Fixed bias (base bias) — the bad example you compare against

Single R_B from V_{CC} to base, R_C to collector. $I_B = (V_{CC} - 0.7)/R_B \Rightarrow I_C = \beta \cdot (V_{CC} - 0.7)/R_B$ — **directly proportional to β** . If β doubles, I_C doubles and the stage may saturate. Mention this circuit in design answers as the motivation for the voltage divider.

Adding R_E (emitter bias / fixed bias with R_E , §1.3 example) helps via negative feedback: $I_C = \beta(V_{CC}-0.7)/[R_B+(\beta+1)R_E]$ — better, still β -dependent unless $(\beta+1)R_E \gg R_B$.

C. Voltage-divider bias — THE β -independent circuit ●



Voltage-divider bias, CE form with R_C (left) and common-collector form with the collector tied to V_{CC} (right). Thévenin base network: $V_{BB} = V_{CC} \cdot R_2 / (R_1 + R_2)$, $R_{Th} = R_1 \parallel R_2$.

Exam diagram — practice drawing this (with coupling/bypass capacitors added when the question says "amplifier": C_{in} at base, C_{out} at collector, C_E across R_E).

Exact analysis (Thévenin): base loop KVL: $V_{BB} = I_B \cdot R_{Th} + V_{BE} + I_E \cdot R_E$, with $I_B = I_E / (\beta + 1)$:

$$I_E = (V_{BB} - V_{BE}) / [R_E + R_{Th} / (\beta + 1)]$$

β -independence condition: if $R_{Th} / (\beta + 1) \ll R_E$, i.e. $R_{Th} \ll (\beta + 1)R_E$, then

$$I_E \approx (V_{BB} - V_{BE}) / R_E \text{ — no } \beta \text{ anywhere. } \checkmark$$

Significance of β -independency HOT PYQ 2082 Bhadra Q1(a) [1 mark]:

β varies widely between devices and with temperature. If the divider is designed so that $R_{Th} \ll (\beta + 1)R_E$ (divider current $\gg$ base current), then V_B is fixed by R_1 – R_2 alone and $I_E \approx (V_B - V_{BE}) / R_E$ is set by resistors only. The Q-point then stays put when the transistor is replaced or heats up $\rightarrow$ stable, reproducible, mass-producible amplifier.

Physical stabilizing mechanism (worth 1-2 marks of explanation): V_B is held stiff by the divider. If I_C tries to rise ($\beta \uparrow$ or $T \uparrow$), $V_E = I_E \cdot R_E$ rises, so $V_{BE} = V_B - V_E$ **falls**, throttling injection and pulling I_C back down — **negative feedback via R_E** .

Stiff / firm biasing rules (the design criterion the PYQs name):

Term (as used in your PYQs / Floyd-Boylestad)	Criterion	Meaning
Stiff voltage divider	$R_2 \leq \beta \cdot R_E / 10$ (equivalently $\beta \cdot R_E \geq 10 \cdot R_2$, i.e. base loading $\leq \sim 10\%$)	divider current $\approx 10 \times I_B$; V_B essentially independent of β
Firm biasing	same practical rule, $R_2 \approx \beta \cdot R_E / 10$ (some texts allow up to $\beta \cdot R_E / 5$)	"firmly held" base voltage; treat as the same design step in exams

If the examiner writes "use firm/stiff biasing method," they want you to **choose $R_2 = \beta \cdot R_E / 10$** and then compute R_1 — say so explicitly for the method mark.

D. • Standard design recipe (memorize this — it answers a guaranteed question)

Given: V_{CC} , desired I_C (or I_E), β . **Find:** R_E , R_C (if CE), R_1 , R_2 .

- Emitter voltage:** choose $V_E = V_{CC} / 10$ (rule of thumb: ~ 1 – 2 V; large enough to swamp V_{BE} drift, small enough not to eat swing). $\rightarrow R_E = V_E / I_E$ ($I_E \approx I_C \cdot (\beta + 1) / \beta \approx I_C$).

2. **Q at mid-line for max swing:** choose $V_{CE} = V_{CC}/2 \rightarrow V_C = V_{CE} + V_E \rightarrow R_C = (V_{CC} - V_C)/I_C$. (CC amplifier: no R_C — instead put $V_E = V_{CC}/2$, see Design 3.)
3. **Base voltage:** $V_B = V_E + 0.7 \text{ V}$.
4. **Stiff/firm divider:** $R_2 = \beta \cdot R_E / 10$.
5. **R_1 from the divider ratio:** $V_B = V_{CC} \cdot R_2 / (R_1 + R_2) \Rightarrow R_1 = R_2 \cdot (V_{CC} - V_B) / V_B$.
6. (State the check: $R_{Th} = R_1 \parallel R_2 \ll (\beta + 1)R_E \checkmark$.)

(Alternative guideline — Sedra-Smith "one-third rule": drop $V_{CC}/3$ across R_C , $V_{CC}/3$ across V_{CE} , $V_{CC}/3$ across R_E . Acceptable if you state it; the $V_{CC}/10 + V_{CC}/2$ recipe above matches IOE marking schemes.)

E. HOT Worked PYQ Designs — all four years

Design 1 — PYQ 2082 Bhadra Q1(b) [5]: β -independent voltage-divider bias; $V_{CC} = 15 \text{ V}$, $I_E = 1 \text{ mA}$, $\beta = 100$.

1. $V_E = 15/10 = 1.5 \text{ V} \rightarrow R_E = 1.5 \text{ V} / 1 \text{ mA} = 1.5 \text{ k}\Omega$
2. $I_C = \alpha I_E = (100/101)(1 \text{ mA}) \approx 0.99 \text{ mA} \approx 1 \text{ mA}$. $V_{CE} = 7.5 \text{ V} \rightarrow V_C = 7.5 + 1.5 = 9 \text{ V} \rightarrow R_C = (15 - 9)/1 \text{ mA} = 6 \text{ k}\Omega$
3. $V_B = 1.5 + 0.7 = 2.2 \text{ V}$
4. $R_2 = \beta R_E / 10 = 100 \times 1.5\text{k}/10 = 15 \text{ k}\Omega$
5. $R_1 = 15\text{k} \times (15 - 2.2)/2.2 = 15\text{k} \times 12.8/2.2 \approx 87.3 \text{ k}\Omega$ ($\approx 82\text{--}91 \text{ k}\Omega$ standard value; state nearest standard)
6. Check: $R_{Th} = 87.3\text{k} \parallel 15\text{k} \approx 12.8 \text{ k}\Omega \ll (\beta + 1)R_E = 151.5 \text{ k}\Omega \checkmark$ β -independent.

Design 2 — PYQ 2081 Ashwin Q1 [4]: β -independent CE amplifier, stiff biasing; $V_{CC} = 20 \text{ V}$, $I_C = 1.5 \text{ mA}$, $\beta = 110$.

1. $I_E = I_C(\beta + 1)/\beta = 1.5 \times 111/110 \approx 1.51 \text{ mA}$. $V_E = 2 \text{ V} \rightarrow R_E = 2/1.51\text{m} \approx 1.32 \text{ k}\Omega$ ($\approx 1.3 \text{ k}\Omega$)
2. $V_{CE} = 10 \text{ V} \rightarrow V_C = 12 \text{ V} \rightarrow R_C = (20 - 12)/1.5\text{m} = 5.33 \text{ k}\Omega$ ($\approx 5.6 \text{ k}\Omega$ std)
3. $V_B = 2.7 \text{ V}$
4. Stiff: $R_2 = \beta R_E / 10 = 110 \times 1.32\text{k}/10 \approx 14.6 \text{ k}\Omega$ ($\approx 15 \text{ k}\Omega$)
5. $R_1 = 14.6\text{k} \times (20 - 2.7)/2.7 \approx 93.5 \text{ k}\Omega$ ($\approx 91 \text{ k}\Omega$ std)

Design 3 — PYQ 2083 Baishakh Q1(b) [5]: β -independent DC-biased common-collector amplifier, firm biasing; $V_{CC} = 20 \text{ V}$, $I_C = 2 \text{ mA}$, $\beta = 100$.

Key difference: CC (emitter follower) $\rightarrow$ **collector tied directly to V_{CC} (no R_C)**; output taken at emitter; $V_{CE} = V_{CC} - V_E$.

1. For maximum symmetric emitter swing choose $V_E = V_{CC}/2 = 10 \text{ V}$ (so $V_{CE} = 10 \text{ V}$, Q mid-line).
2. $I_E = I_C(\beta + 1)/\beta = 2 \times 101/100 = 2.02 \text{ mA} \approx 2 \text{ mA} \rightarrow R_E = 10 \text{ V}/2.02 \text{ mA} \approx 4.95 \text{ k}\Omega \approx 5 \text{ k}\Omega$ (4.7 k Ω std)
3. $V_B = V_E + 0.7 = 10.7 \text{ V}$
4. Firm: $R_2 = \beta R_E / 10 = 100 \times 5\text{k}/10 = 50 \text{ k}\Omega$
5. $R_1 = 50\text{k} \times (20 - 10.7)/10.7 = 50\text{k} \times 9.3/10.7 \approx 43.5 \text{ k}\Omega$ (43 k Ω / 47 k Ω std)
6. Check: $R_{Th} \approx 23.3 \text{ k}\Omega \ll (\beta + 1)R_E \approx 505 \text{ k}\Omega \checkmark$. Draw the circuit: divider R_1/R_2 at base, collector $\rightarrow V_{CC}$, R_E to ground, output via capacitor from emitter.

Design 4 — PYQ 2082 Baishakh Q1 [4]: β -independent voltage-divider bias; $V_{CC} = 12 \text{ V}$, $I_E = 1 \text{ mA}$, $\beta = 100$.

1. $V_E = 1.2 \text{ V} \rightarrow R_E = 1.2 \text{ k}\Omega$
2. $V_{CE} = 6 \text{ V} \rightarrow V_C = 7.2 \text{ V} \rightarrow R_C = (12 - 7.2)/1 \text{ mA} = 4.8 \text{ k}\Omega$ (4.7 k Ω std)
3. $V_B = 1.9 \text{ V}$

4. $R_2 = 100 \times 1.2\text{k}/10 = 12\text{ k}\Omega$

5. $R_1 = 12\text{k} \times (12 - 1.9)/1.9 \approx 63.8\text{ k}\Omega$ (62 k Ω / 68 k Ω std)

Marking-scheme observation: every design carries the same 5 mark-points — (i) circuit diagram, (ii) V_E & R_E choice with justification, (iii) $V_{CE} = V_{CC}/2$ & R_C (or $V_E = V_{CC}/2$ for CC), (iv) stiff/firm R_2 rule stated and used, (v) R_1 from divider equation. Write the rules as *sentences* before substituting numbers.

F. Common mistakes ●

- Putting R_C in a **common-collector** design (Design 3) — instant concept error.
- Using I_C and I_E interchangeably without stating $I_E \approx I_C$.
- Computing R_1 first: you can't — **R_2 comes from the stiffness rule first**, then R_1 from the ratio.
- Forgetting $V_B = V_E + 0.7$ (not V_E).
- Not stating the stiff/firm criterion when the question explicitly names it — that phrase is worth a mark.

FORMULAS Formula Box — §1.6

Formula	Role
$V_{BB} = V_{CC} \cdot R_2 / (R_1 + R_2)$, $R_{Th} = R_1 \parallel R_2$	Thévenin base network
$I_E = (V_{BB} - V_{BE}) / [R_E + R_{Th} / (\beta + 1)]$	Exact bias current
Condition $R_{Th} \ll (\beta + 1)R_E \Rightarrow I_E \approx (V_{BB} - 0.7) / R_E$	β -independence
Design set: $V_E = V_{CC}/10$; $V_{CE} = V_{CC}/2$; $V_B = V_E + 0.7$; $R_2 = \beta R_E / 10$; $R_1 = R_2(V_{CC} - V_B) / V_B$	The recipe

PYQ Check (HOT REPEATED PYQ TOPIC — every year)

- 2083 Baishakh Q1 [1+5] — CC design, firm bias (Design 3)
- 2082 Bhadra Q1 [1+5] — significance of β -independency + design (Design 1)
- 2082 Baishakh Q1 [4] — design (Design 4)
- 2081 Ashwin Q1 [4] — CE design, stiff bias (Design 2)

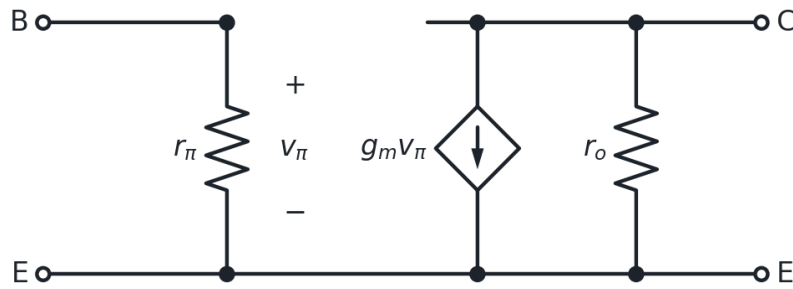
Exam-ready when: given any (V_{CC} , I_C or I_E , β , configuration), you produce circuit + all four resistors + stated rules in under 8 minutes.

1.7 Small-Signal Equivalent Circuit Models (π and T)

A. What a small-signal model is

For AC analysis we (i) kill DC sources ($V_{CC} \rightarrow$ AC ground), (ii) short coupling/bypass capacitors, (iii) **replace the transistor by a linear circuit** valid for small wiggles about Q. Two equivalent linear circuits exist — π and T. Same device, same equations, different convenience.

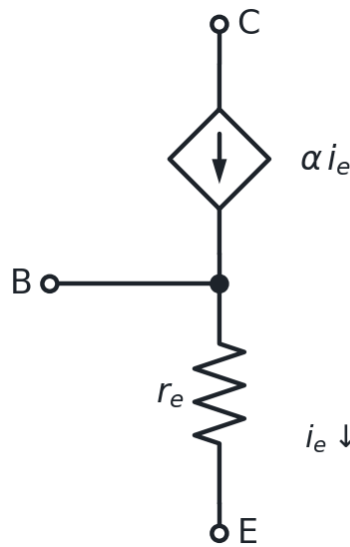
B. The hybrid- π model ● (best for CE)



Hybrid- π small-signal model: r_π from B to E carrying i_b , dependent source $g_m \cdot v_\pi (\equiv \beta \cdot i_b)$ from C to E, in parallel with $r_o = V_A/I_C$.

- Input: r_π between B-E carrying i_b , with $v_\pi = v_{be}$ across it. - Output: dependent current source $g_m \cdot v_\pi$ (or $\beta \cdot i_b$) from C to E, in parallel with $r_o = V_A/I_C$ (include r_o only when given/needed).

C. The T model (best for CB and CC)



T model: $r_e = V_T/I_E$ in the emitter leg carrying the full i_e ; collector source $\alpha \cdot i_e$. Looking into the base you see $(\beta+1)(r_e + \text{emitter-path})$ — the reflection rule.

- $r_e = V_T/I_E$ in series with the emitter; collector source $\alpha \cdot i_e \approx i_e$. - Looking into the **base** of the T model you see $(\beta+1) \cdot (r_e + \text{anything in the emitter path})$ — the **resistance-reflection rule**, the workhorse of CC/CE-with- R_E analysis.

D. Model selection & equivalence

	Hybrid- π	T model
Parameters	r_π, g_m (or $\beta \cdot i_b$), r_o	$r_e, \alpha (\approx 1)$
Natural for	CE	CB, CC (emitter follower)
Input resistance element	r_π at base	r_e at emitter
Bridge	$r_\pi = (\beta+1)r_e \approx \beta r_e$; $g_m r_e = \alpha$; $g_m r_\pi = \beta$	

Both give identical answers — choose whichever makes the algebra shortest, and **say which model you are using** in the exam.

KEY Section Summary

- **Draw:** both models from memory. **Exam diagram — practice drawing these.** Every §1.9 derivation begins with one of them; the model diagram itself carries ~1 mark of each 5-mark question.
- **Memorize:** the reflection rule $R_{in}(\text{base}) = (\beta+1)(r_e + R_{\text{emitter-path}})$.

1.8 Basic Single-Stage BJT Amplifier Configurations (C-B, C-E, C-C)

A. Naming rule

The configuration is named after the terminal that is **common to input and output (AC ground)**: input-output pairs are E→C (Common-**B**ase), B→C (Common-**E**mitter), B→E (Common-**C**ollector / emitter follower).

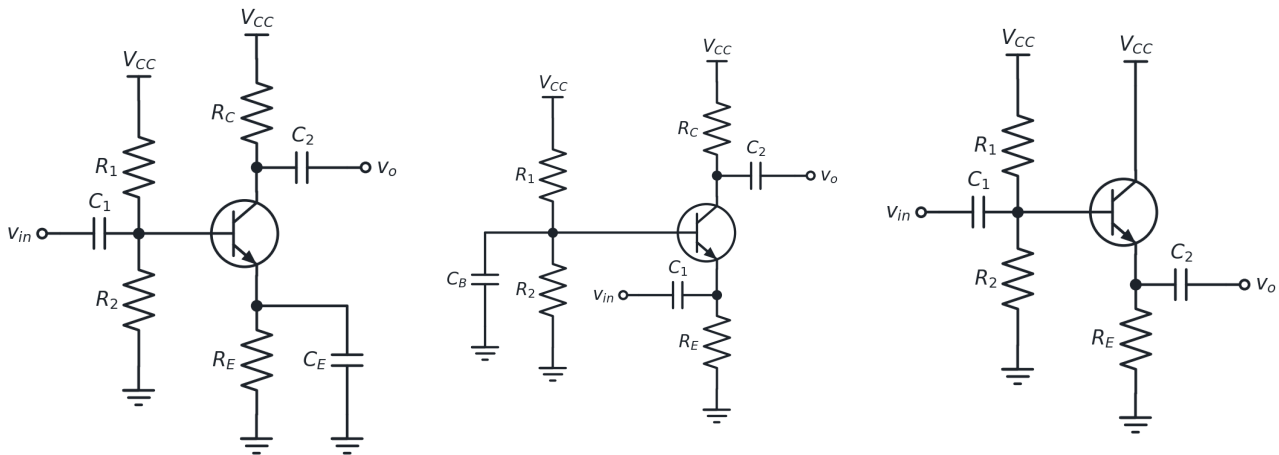
B. **HOT** Role of coupling capacitors — PYQ 2082 Bhadra Q2(a) [2 marks]

Coupling capacitors (C_1 at input, C_2 at output) **pass the AC signal while blocking DC**. They connect source→amplifier→load without letting the source or load resistance disturb the DC Q-point (which would shift bias), and without letting the amplifier's DC voltages appear across the load. At signal (mid-band) frequencies $X_C = 1/(2\pi fC)$ is negligible → the capacitor is an effective **short for AC, open for DC**. Side-effect: they set the amplifier's **lower cutoff frequency** (blocking very low frequencies/DC).

C. **HOT** Role of the bypass capacitor — PYQ 2083 Baishakh Q2(a) [2 marks]

The bypass capacitor C_E is placed **in parallel with R_E** . At DC it is open, so R_E still provides the negative feedback that stabilizes the Q-point (§1.6). At signal frequencies it is a short, **bypassing R_E to ground** so the emitter is AC-grounded. Without it the gain collapses from $|A_v| = R_C/r_e$ to $R_C/(r_e + R_E)$ (§1.9-C). So C_E lets you keep **DC bias stability AND high AC gain simultaneously**. It also raises input-side gain by removing R_E from the reflected base impedance.

D. The three circuits (AC signal paths)



The three single-stage configurations with divider bias: CE with bypassed R_E (left), CB with base AC-grounded via C_B and input at the emitter (middle), CC/emitter follower with the collector tied to V_{CC} and output at the emitter (right).

Exam diagrams — practice drawing all three with the divider bias from §1.6 attached.

E. • Master comparison table (memorize — a compare question is always possible)

Property	Common-Emitter	Common-Base	Common-Collector (Emitter Follower)
Input / output terminals	B / C	E / C	B / E
Input impedance	Medium ($\approx R_1 \parallel R_2 \parallel \beta r_e$, $\sim k\Omega$)	Low ($\approx r_e$, $\sim$ tens of Ω)	High ($((\beta+1)(r_e + R_E \parallel R_L))$, $\sim 100\text{ k}\Omega$)

Property	Common-Emitter	Common-Base	Common-Collector (Emitter Follower)
Output impedance	Medium-high ($\approx R_C$)	High ($\approx R_C$)	Low ($\approx r_e + R_S/(\beta+1)$)
Voltage gain A_v	High , inverting ($-g_m R_C$)	High, non-inverting ($+g_m R_C$)	≈ 1 (slightly less), non-inverting
Current gain A_i	High ($\approx \beta$)	$\approx \alpha \approx 1$	High ($\approx \beta+1$)
Phase shift	180°	0°	0°
Typical use	General voltage amplifier	High-frequency stages, current buffer	Buffer / impedance matcher , output stage

1.9 Small-Signal Analysis of Amplifiers HOT HOT (GUARANTEED QUESTION — 4/4 YEARS)

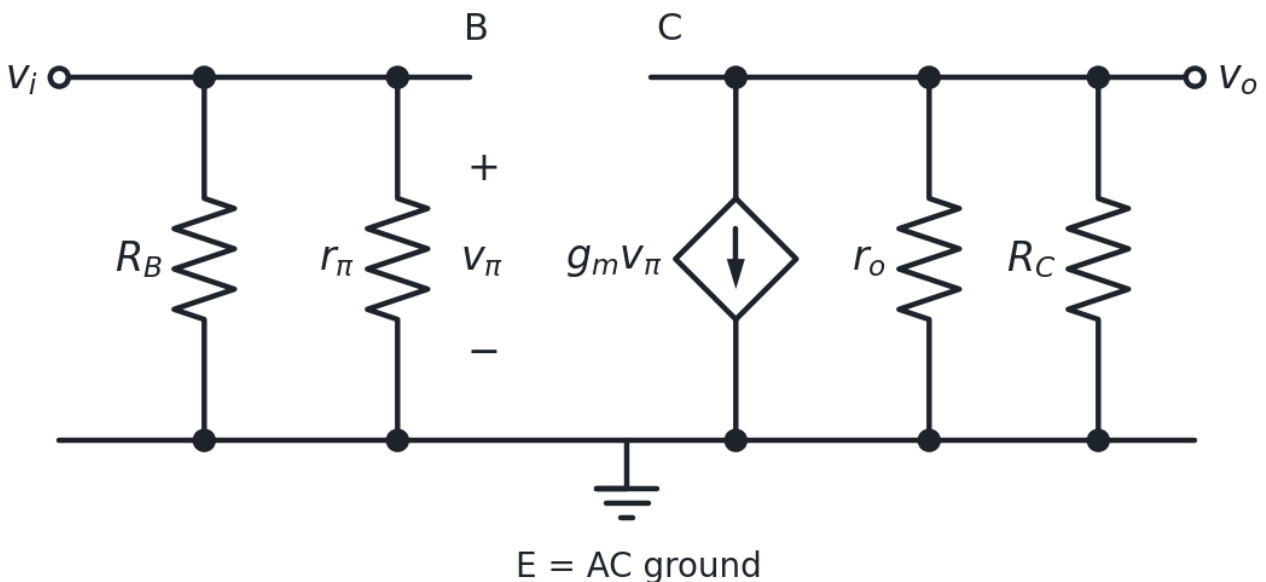
A. Universal 4-step method (state it, then execute it)

1. **DC analysis** $\rightarrow I_E$ (Q-point) $\rightarrow$ compute r_e ($= 26 \text{ mV}/I_E$), r_π ($= \beta r_e$), g_m ($= I_C/V_T$).
2. **Draw the AC equivalent:** $V_{CC} \rightarrow$ ground; capacitors $\rightarrow$ shorts; then substitute the π or T model.
(This drawing is explicitly demanded by every PYQ — never skip it.)
3. Write KVL/KCL for $\mathbf{Z_i = v_i/i_i}$, $\mathbf{Z_o = v_o/i_o}$ (with $\mathbf{v_{in} = 0}$), $\mathbf{A_v = v_o/v_i}$ (and A_i when asked).
4. Sanity-check signs and magnitudes against the §1.8 table.

Below, $R_B \equiv R_1 \parallel R_2$ (the bias divider as seen by AC).

B. HOT Common-Emitter amplifier (bypassed R_E) — PYQ 2082 Bhadra Q2(b) [5]

AC equivalent (hybrid- π): source $v_i \rightarrow$ base node; r_π from B to ground (E is AC ground via C_E); R_B from B to ground; at the collector, current source $g_m \cdot v_\pi$ (downward C $\rightarrow$ E) in parallel with r_o and R_C to ground.



Small-signal (AC) equivalent of the bypassed- R_E common-emitter amplifier: $Z_i = R_B \parallel r_\pi$, $Z_o = R_C \parallel r_o$, $A_v = -g_m(R_C \parallel r_o)$.

Input impedance. Looking into the base: only r_π to ground (emitter grounded). With the divider:

$$\mathbf{Z_i = R_B \parallel r_\pi = R_1 \parallel R_2 \parallel \beta r_e}$$

Output impedance. Set $v_i = 0 \Rightarrow v_\pi = 0 \Rightarrow$ dependent source = 0 (open). Looking back from the output:

$$Z_o = R_C \parallel r_o \approx R_C \text{ (when } r_o \geq 10R_C \text{)}$$

Voltage gain. $v_{\pi} = v_i$ (base to grounded emitter). Output node: $v_o = -(g_m v_{\pi})(R_C \parallel r_o)$:

$$A_v = -g_m(R_C \parallel r_o) = -(R_C \parallel r_o)/r_e \approx -R_C/r_e \text{ (negative } \Rightarrow 180^\circ \text{ inversion)}$$

With an external load R_L (coupled by C_2): replace R_C by $R_C \parallel R_L$.

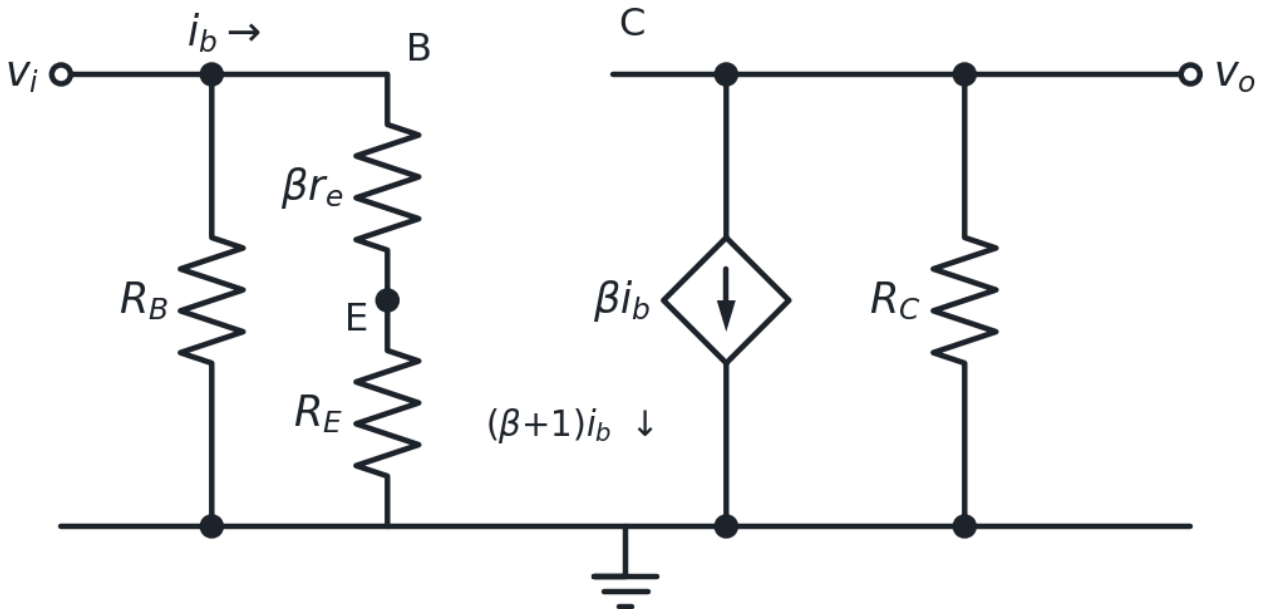
(Current gain if asked: $A_i = -A_v Z_i / R_C \approx \beta R_B / (R_B + r_{\pi}) \approx \beta$ for stiff R_B .)

C. **HOT** CE with UNBYPASSED R_E + Worked numerical — PYQ 2082 Baishakh Q2 [5]

Given circuit: $V_{CC} = 20 \text{ V}$; $R_B = 470 \text{ k}\Omega$ (single base resistor); $R_C = 2.2 \text{ k}\Omega$; $R_E = 0.56 \text{ k}\Omega$ (no bypass capacitor shown); C_1, C_2 coupling; $\beta = 120$, $r_o = 40 \text{ k}\Omega$. Find Z_i , A_v , Z_o .

Step 1 — DC (done in §1.3-B): $I_B = 35.9 \mu\text{A}$, $I_E = 4.34 \text{ mA} \Rightarrow r_e = 26 \text{ mV} / 4.34 \text{ mA} \approx 5.99 \Omega$.

Step 2 — AC equivalent: base sees R_B to ground and, into the base, $r_{\pi} (= \beta r_e)$ in series with the unbypassed R_E (T-model thinking). Collector: $\beta \cdot i_b$ source with R_C ; r_o sits from C to E.



AC equivalent of the CE stage with unbypassed R_E : the base sees $Z_b = \beta r_e + (\beta+1)R_E$ (reflection rule), so $A_v \approx -R_C/(r_e + R_E)$.

Step 3 — key derivation (reflection rule). Base current i_b flows in r_{π} ; emitter carries $(\beta+1)i_b$ through R_E : $v_i = i_b \cdot \beta r_e + (\beta+1)i_b \cdot R_E \Rightarrow$

$$Z_b = v_i / i_b = \beta r_e + (\beta+1)R_E \approx \beta(r_e + R_E) \quad Z_i = R_B \parallel Z_b; \quad Z_o \approx R_C; \quad A_v = -\beta R_C / Z_b \approx -R_C / (r_e + R_E)$$

(These approximations require $r_o \geq 10(R_C + R_E)$ — that is exactly why the question gives you r_o : to check it.)

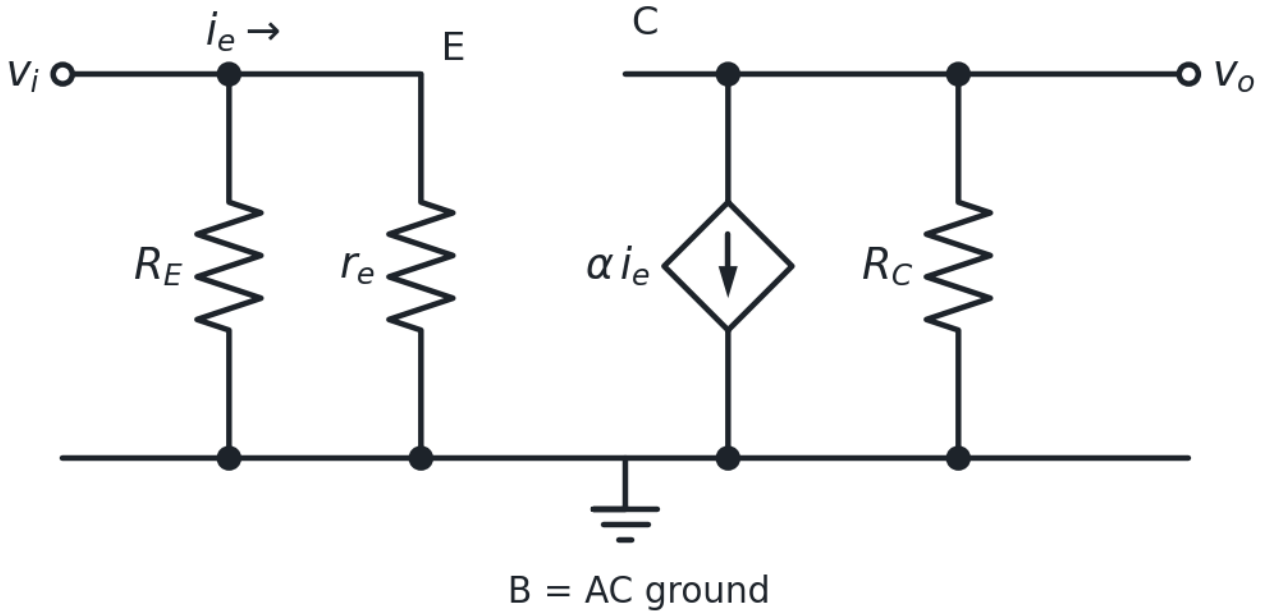
Step 4 — numbers. - $Z_b = 120 \times 5.99 + 121 \times 560 \approx 719 + 67,760 \Rightarrow \approx 67.9 \text{ k}\Omega$ (dominated by the reflected R_E) - $Z_i = 470 \text{ k}\Omega \parallel 67.9 \text{ k}\Omega \approx 59.3 \text{ k}\Omega$ - Check r_o : $10(R_C + R_E) = 10(2.2\text{k} + 0.56\text{k}) = 27.6 \text{ k}\Omega \leq r_o = 40 \text{ k}\Omega \checkmark \Rightarrow r_o$ negligible. - $Z_o \approx R_C = 2.2 \text{ k}\Omega$ - $A_v = -\beta R_C / Z_b = -(120 \times 2.2\text{k}) / 67.9\text{k} \approx -3.89 \Rightarrow$ small gain, inverted — the price of an unbypassed R_E (this is the numerical proof of the bypass-capacitor story in §1.8-C).

Exam shortcut / observation: when $(\beta+1)R_E \gg \beta r_e$, $A_v \approx -R_C / R_E = -2.2\text{k} / 0.56\text{k} \approx -3.93$ — a 5-second sanity check of your long answer.

D. HOT Common-Base amplifier — PYQ 2083 Baishakh Q2(b) [5]

Circuit: input at emitter through C_1 ; base AC-grounded via base bypass capacitor C_B (divider R_1 – R_2 still sets DC); output at collector; R_E from emitter to ground, R_C at collector.

AC equivalent (T model is natural):



AC equivalent of the common-base amplifier (T model): $Z_i = R_E \parallel r_e \approx r_e$, $Z_o \approx R_C$, $A_v = +\alpha \cdot R_C / r_e$ — non-inverting.

Input impedance. Looking into the emitter (base grounded): $v_i = i_e \cdot r_e \Rightarrow R_{in}(\text{emitter}) = r_e$. With R_E in parallel:

$$Z_i = R_E \parallel r_e \approx r_e \text{ (very low, tens of } \Omega \text{ — the defining CB feature)}$$

Output impedance. $v_i = 0 \Rightarrow i_e = 0 \Rightarrow \text{source } \alpha \cdot i_e = 0$ (open):

$$Z_o = R_C \text{ (in parallel with the very large CB output resistance of the transistor } \approx r_o(1 + g_{mR-E'}) \text{ — state } Z_o \approx R_C)$$

Voltage gain. $i_e = v_i / r_e$; $v_o = +(\alpha \cdot i_e)(R_C \parallel R_L)$:

$$A_v = +\alpha \cdot R_C / r_e \approx +g_m \cdot R_C = +R_C / r_e \text{ — same magnitude as CE but non-inverting (0°)}$$

Current gain: $A_i = i_c / i_e = \alpha \approx 1$ (slightly less than unity — CB is not a current amplifier).

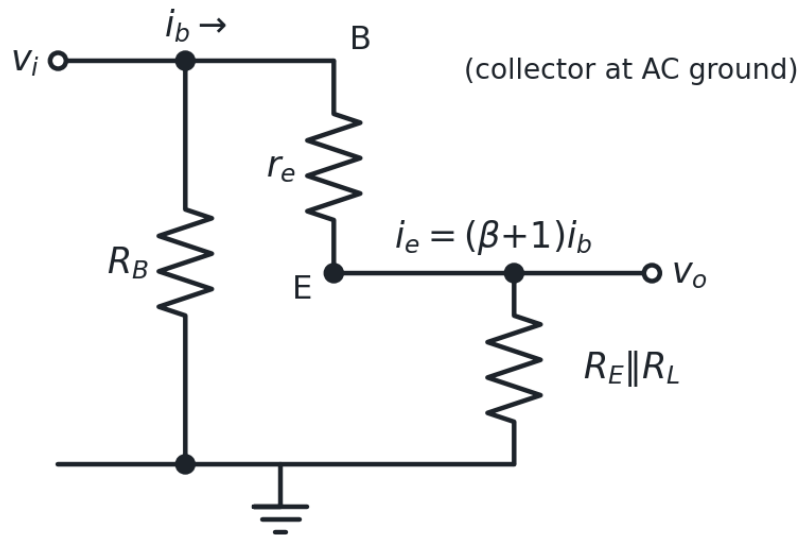
E. HOT Common-Collector (Emitter Follower) — PYQ 2081 Ashwin Q2 [1+4]

Part (a) — Why "emitter follower"? [1]:

The output is taken at the emitter, and $v_E = v_B - v_{BE}$ with v_{BE} nearly constant (~ 0.7 V DC, tiny signal variation). Hence the emitter voltage **follows the base voltage** almost exactly: $A_v = v_o / v_i \approx 1$, in phase. The stage gives no voltage gain but large current gain and an impedance step-down — the transistor equivalent of a buffer.

Part (b) — model + derivations [4]. Circuit: divider bias at base, collector directly to V_{CC} (AC ground!), R_E emitter→ground, output across R_E ($\parallel R_L$ via C_2).

AC equivalent (T model):



AC equivalent of the emitter follower: collector at AC ground, output across $R_E \parallel R_L$; $R_{in}(\text{base}) = (\beta + 1)(r_e + R_E)$, $A_v = R_E / (r_e + R_E) \approx 1$.

(i) Input resistance. $v_i = i_b \cdot (\beta + 1)(r_e + R_E)$ where $R_E = R_E \parallel R_L$ (reflection rule):

$$R_{in}(\text{base}) = (\beta + 1)(r_e + R_E); Z_i = R_B \parallel (\beta + 1)(r_e + R_E) \text{ — high (often } \gg 100 \text{ k}\Omega)$$

(ii) Voltage gain. The emitter current i_e flows through r_e then R_E ; output is across R_E — a voltage divider: $v_i = i_e(r_e + R_E)$, $v_o = i_e R_E \Rightarrow$

$$A_v = R_E / (r_e + R_E) \approx 1 \text{ (but } < 1), \text{ non-inverting — because } r_e \text{ (a few } \Omega) \ll R_E \text{ (k}\Omega)$$

(iii) Current gain. $i_e = (\beta + 1)i_b \Rightarrow$ device current gain = $\beta + 1$. Overall (including the R_B divider): $A_i = (\beta + 1) \cdot R_B / (R_B + Z_b) \approx \beta + 1$ for large R_B . State both.

(iv) Output impedance (know it — often the follow-up): kill v_i , look into the emitter; the source resistance $R_s = R_s \parallel R_B$ seen at the base gets **divided** by $(\beta + 1)$:

$$Z_o = R_E \parallel [r_e + R_s / (\beta + 1)] \approx r_e + R_s / (\beta + 1) \text{ — very low } (\Omega\text{-range}) \Rightarrow \text{ideal buffer/output stage.}$$

F. • Results comparison table (the one-look revision grid)

Quantity	CE (bypassed)	CE (unbypassed R_E)	CB	CC
Z_i	$R_B \parallel \beta r_e$	$R_B \parallel \beta(r_e + R_E)$	$R_E \parallel r_e \approx r_e$	$R_B \parallel (\beta + 1)(r_e + R_E)$
Z_o	$R_C \parallel r_o \approx R_C$	$\approx R_C$	$\approx R_C$	$r_e + R_s / (\beta + 1)$
A_v	$-R_C / r_e$ (high, inv.)	$-R_C / (r_e + R_E)$ (low, inv.)	$+R_C / r_e$ (high, non-inv.)	$R_E / (r_e + R_E) \approx 1$
A_i	$\approx \beta$	$\approx \beta$	$\approx \alpha \approx 1$	$\approx \beta + 1$

(Primes denote $\parallel$ with load: $R_C' = R_C \parallel R_L$, $R_E' = R_E \parallel R_L$.)

G. Common mistakes •

- Skipping the DC step — r_e is unknown without I_E , and the whole answer collapses.
- **Not drawing the AC equivalent** when the question says "draw the small-signal equivalent circuit" — that's ~2 of the 5 marks.
- Dropping the $(\beta + 1)$ factor in the reflection rule, or reflecting the wrong way for Z_o of the follower (divide by $(\beta + 1)$, don't multiply).
- Sign of A_v : CE negative; CB and CC positive.

- Forgetting to check/mention $r_o \geq 10(R_C + R_E)$ when r_o is given (2082 Baishakh Q2 gives r_o precisely to test this).

PYQ Check (HOT REPEATED PYQ TOPIC — every year, configuration rotates)

- 2082 Bhadra Q2 [2+5] — coupling caps + CE derivation (B above)
- 2082 Baishakh Q2 [5] — CE-with- R_E numerical (C above, full solution)
- 2083 Baishakh Q2 [2+5] — bypass cap + CB derivation (D above)
- 2081 Ashwin Q2 [1+4] — emitter follower + CC derivations (E above)

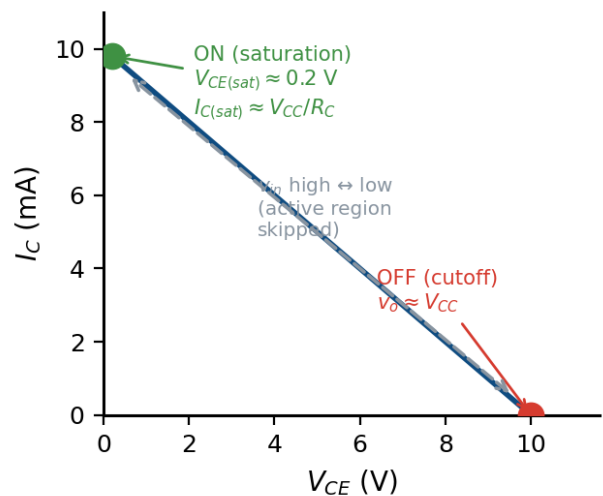
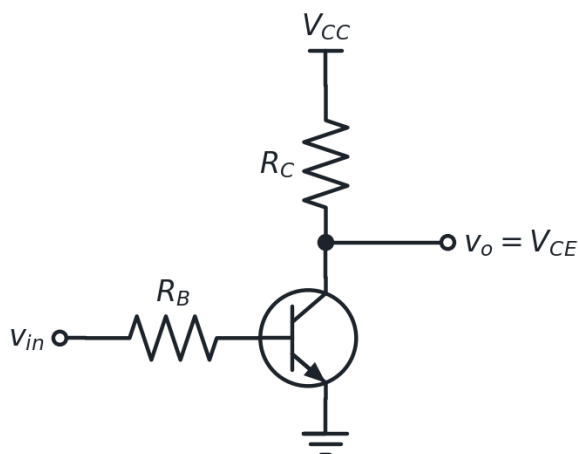
Exam-ready when: for each configuration you can, from a blank page: draw circuit → draw AC model → derive Z_i , Z_o , A_v (A_i) → quote typical magnitudes — in ≤ 10 minutes.

1.10 Transistor as a Switch — Cutoff and Saturation • PYQ 2082 Baishakh Q3 [4]

A. Concept

Restrict operation to the two **ends of the load line** — cutoff (open switch) and saturation (closed switch) — skipping the active region entirely. This is the basis of BJT logic/driver circuits (the circuit is a logic **inverter**).

B. Circuit and the two states



BJT switch (logic inverter): circuit (left) and its two operating points at the ends of the load line — ON in saturation, OFF in cutoff, the active region skipped (right).

Exam diagram — practice drawing both (circuit + load line with the two operating points marked).

State 1 — Cutoff (switch OPEN): $v_{in} \approx 0$ (< 0.5 V) → EBJ not forward → $I_B = 0$ → $I_C \approx 0$, $v_o = V_{CE} \approx V_{CC}$. The C-E path behaves as an open switch (only leakage I_{CEO} flows).

State 2 — Saturation (switch CLOSED): v_{in} high → I_B large. Maximum possible collector current is fixed by the external circuit:

$$I_{C(sat)} = (V_{CC} - V_{CE(sat)})/R_C \approx V_{CC}/R_C, \text{ with } V_{CE(sat)} \approx 0.2 \text{ V}$$

Condition to guarantee saturation:

$I_B > I_{C(sat)}/\beta$ (design with overdrive: $I_B \approx 2\text{--}10\times$ this minimum; equivalently the circuit forces $\beta I_B > I_{C(sat)}$, so the transistor cannot deliver βI_B and both junctions become forward-biased)

The C-E path then behaves as a closed switch with only ~ 0.2 V across it.

C. 4-mark answer skeleton

(i) circuit diagram; (ii) cutoff explanation with $v_o \approx V_{CC}$; (iii) saturation explanation with $I_{C(sat)}$, $V_{CE(sat)}$, and the $I_B > I_{C(sat)}/\beta$ condition; (iv) load-line/inverter remark. Add one number example if time permits (e.g., $V_{CC} = 10$ V, $R_C = 1$ k $\Omega \Rightarrow I_{C(sat)} \approx 9.8$ mA; $\beta = 100 \Rightarrow$ need $I_B > 98$ μ A).

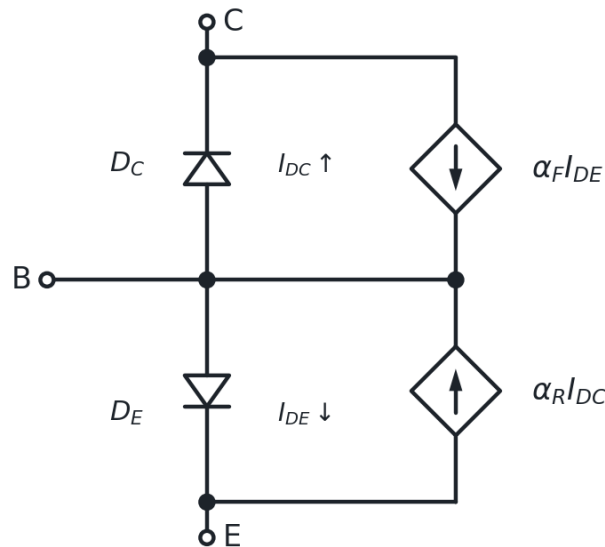
Common mistakes

- Writing $I_C = \beta I_B$ **in saturation** — false; in saturation $I_C < \beta I_B$ and I_C is set by R_C .
- Using $V_{CE} = 0$ exactly instead of $V_{CE(sat)} \approx 0.2$ V.

1.11 A General Large-Signal Model of the BJT: The Ebers-Moll Model • (syllabus-complete; no PYQ in your 4 papers)

A. Idea

One model valid in **all four modes**: represent the BJT as its two physical diodes **plus** two dependent sources expressing the coupling (transistor action) between them.



Ebers-Moll model: each junction is a diode (D_E , D_C), and each dependent source injects a fraction (α_F forward, α_R reverse) of the other junction's diode current.

B. Equations (npn)

Diode currents: $I_{DE} = I_{ES}(e^{V_{BE}/V_T} - 1)$, $I_{DC} = I_{CS}(e^{V_{BC}/V_T} - 1)$. Terminal currents:

$$I_E = I_{ES}(e^{V_{BE}/V_T} - 1) - \alpha_R I_{CS}(e^{V_{BC}/V_T} - 1) \quad I_C = \alpha_F I_{ES}(e^{V_{BE}/V_T} - 1) - I_{CS}(e^{V_{BC}/V_T} - 1) \quad \text{Reciprocity:} \\ \alpha_F I_{ES} = \alpha_R I_{CS} = I_S$$

Symbol	Meaning	Typical
α_F	Forward CB gain (normal α)	0.98–0.998
α_R	Reverse CB gain (poor — asymmetric doping)	0.02–0.5
I_{ES} , I_{CS}	EBJ / CBJ diode scale currents	fA–pA

C. Modes recovered as special cases

- **Active:** $V_{BE} > 0$, $V_{BC} < 0 \Rightarrow$ second exponentials $\approx 0 \Rightarrow I_C \approx \alpha_F I_{ES} e^{V_{BE}/V_T} = I_S e^{V_{BE}/V_T}$ — recovers §1.1.
- **Saturation:** both exponentials active $\Rightarrow V_{CE(sat)} = V_{BE} - V_{BC} \approx 0.2$ V emerges naturally.

- **Cutoff:** both $\approx -(\text{leakage}) \Rightarrow \text{currents} \approx 0$.

Exam-ready when: you can draw the two-diode/two-source model, write the two equations + reciprocity, and show how active mode falls out. Prepare at 4-mark depth; probability low but it is *the only Chapter-1 topic never yet asked* — a fresh-question candidate.

BJT Final Revision

1. One-Shot Conceptual Summary

A BJT is two junctions sharing a thin, lightly doped base. Forward-bias the EBJ and reverse-bias the CBJ (**active mode**) and the heavily doped emitter injects carriers that diffuse across the thin base and are collected: $I_C = I_S \cdot e^{V_{BE}/V_T} \approx \beta I_B \approx \alpha I_E$ — a voltage-controlled current source using **both carrier types (bipolar)**. Its behaviour is drawn as **output characteristics** (cutoff / active / saturation; slight tilt = Early effect $\Rightarrow r_o = V_A/I_C$). The external collector circuit adds a **load line**; their intersection is the **Q-point**, which we place mid-line ($V_{CE} \approx V_{CC}/2$) for maximum swing using a **voltage-divider bias** designed so $R_{Th} \ll (\beta+1)R_E$ — making $I_E \approx (V_B - 0.7)/R_E$, **independent of β** (stiff/firm rule $R_2 = \beta R_E/10$). Around Q, linearize: $g_m = I_C/V_T$, $r_\pi = \beta/g_m$, $r_e \approx 26 \text{ mV}/I_E$, giving the **hybrid- π** and **T** models. Insert these into the three configurations: **CE** = high inverting voltage gain ($-R_C/r_e$; falls to $-R_C/(r_e + R_E)$ if R_E unbypassed — hence the **bypass capacitor**; **coupling capacitors** isolate DC while passing signal); **CB** = low Z_{in} , non-inverting gain, $A_i \approx 1$; **CC (emitter follower)** = $A_v \approx 1$, huge Z_{in} , tiny Z_{out} — a buffer. Drive the base hard or not at all and the BJT is a **switch**: cutoff (open, $v_o \approx V_{CC}$) $\leftrightarrow$ saturation (closed, $V_{CE(sat)} \approx 0.2 \text{ V}$, need $I_B > I_{C(sat)}/\beta$). The **Ebers-Moll model** (two diodes + two α -coupled sources) unifies all modes in one large-signal description.

2. Important Diagrams (practice each until automatic)

#	Diagram	Where	Priority
D1	nnp structure + npn/pnp symbols with current directions	§1.1	●
D2	CE output characteristics, 3 regions, Early extrapolation	§1.2	● (embedded everywhere)
D3	DC load line with Q-point and clipping	§1.4	●
D4	Voltage-divider bias circuit (+ CC variant, + capacitors)	§1.6	● GUARANTEED
D5	Hybrid- π model; T model	§1.7	● GUARANTEED
D6	CE, CB, CC full amplifier circuits	§1.8	●
D7	AC equivalent of CE / CB / CC	§1.9	● GUARANTEED
D8	BJT switch circuit + two-point load line	§1.10	●
D9	Ebers-Moll two-diode model	§1.11	●

3. Important Derivations

#	Derivation	Section	Status
Dv1	$\beta = \alpha/(1-\alpha)$ and $\alpha = \beta/(\beta+1)$	§1.1-D	Result to remember (1-line derivation)
Dv2	$g_m = I_C/V_T$ from the exponential law	§1.5-B	● Important derivation (asked 2081)

#	Derivation	Section	Status
Dv3	$r_{\pi} = \beta/g_m, r_e = \alpha/g_m$	\$1.5-C	Short derivation
Dv4	$I_E = (V_{BB} - V_{BE})/[R_E + R_{Th}/(\beta+1)] \rightarrow \beta$ -independence condition	\$1.6-C	● Important (justifies every design)
Dv5	CE: $Z_i = R_B \parallel \beta r_e, Z_o \approx R_C, A_v = -g_m(R_C \parallel r_o)$	\$1.9-B	● Important (asked 2082 Bhadra)
Dv6	Reflection rule $Z_b = \beta r_e + (\beta+1)R_E$	\$1.9-C	● Important (asked 2082 Baishakh)
Dv7	CB: $Z_i \approx r_e, A_v = +R_C/r_e$	\$1.9-D	● Important (asked 2083)
Dv8	CC: $R_{in} = (\beta+1)(r_e + R_E), A_v = R_E/(r_e + R_E), A_i \approx \beta+1, Z_o \approx r_e + R_S/(\beta+1)$	\$1.9-E	● Important (asked 2081)
Dv9	Ebers-Moll equations $\rightarrow$ active-mode limit	\$1.11	● Result to remember

4. Formula Sheet (clean)

Device: $I_E = I_C + I_B, I_C = \beta I_B = \alpha I_E, \beta = \alpha/(1-\alpha), \alpha = \beta/(\beta+1), I_C = I_S e^{V_{BE}/V_T}, r_o = V_A/I_C, V_{BE} \approx 0.7 \text{ V}, V_{CE(sat)} \approx 0.2 \text{ V}$

Small-signal parameters: $g_m = I_C/V_T \approx 40 I_C, r_{\pi} = \beta/g_m = \beta V_T/I_C, r_e = \alpha/g_m \approx 26 \text{ mV}/I_E, r_{\pi} = (\beta+1)r_e$

Load line: $I_C = (V_{CC} - V_{CE})/(R_C + R_E)$; ends V_{CC} and $V_{CC}/(R_C + R_E)$

Voltage-divider bias: $V_{BB} = V_{CC} R_2/(R_1 + R_2), R_{Th} = R_1 \parallel R_2, I_E = (V_{BB} - V_{BE})/[R_E + R_{Th}/(\beta+1)] \cdot \beta$ -independent when $R_{Th} \ll (\beta+1)R_E$

Design recipe: $V_E = V_{CC}/10 \rightarrow R_E = V_E/I_E, V_{CE} = V_{CC}/2 \rightarrow R_C = (V_{CC} - V_{CE} - V_E)/I_C, V_B = V_E + 0.7, R_2 = \beta R_E/10, R_1 = R_2(V_{CC} - V_B)/V_B$ (CC: no $R_C, V_E = V_{CC}/2$)

Amplifiers: CE: $Z_i = R_B \parallel \beta r_e, Z_o \approx R_C \parallel r_o, A_v = -(R_C \parallel r_o \parallel R_L)/r_e, CE + R_E: Z_b = \beta(r_e + R_E), A_v \approx -R_C/(r_e + R_E), CB: Z_i \approx r_e, Z_o \approx R_C, A_v = +R_C/r_e, A_i = \alpha, CC: Z_i = R_B \parallel (\beta+1)(r_e + R_E), A_v = R_E/(r_e + R_E) \approx 1, A_i \approx \beta+1, Z_o \approx r_e + (R_S \parallel R_B)/(\beta+1)$

Switch: $I_{C(sat)} = (V_{CC} - V_{CE(sat)})/R_C$ · guarantee: $I_B > I_{C(sat)}/\beta$

Ebers-Moll: $I_E = I_{ES}(e^{V_{BE}/V_T} - 1) - \alpha_R I_{CS}(e^{V_{BC}/V_T} - 1), I_C = \alpha_F I_{ES}(e^{V_{BE}/V_T} - 1) - I_{CS}(e^{V_{BC}/V_T} - 1), \alpha_F I_{ES} = \alpha_R I_{CS} = I_S$

5. High-Priority Concepts (evidence-based)

- HOT Voltage-divider bias design** (stiff/firm) for CE and CC — appeared in **all 4 papers**.
- HOT Small-signal Z_i, Z_o, A_v of CE, CB, CC** — appeared in **all 4 papers**, configuration rotating; be equally fluent in all three.
- HOT Coupling & bypass capacitor roles** — 2 of 4 papers, always as the 2-mark lead-in.
- g_m derivation · BJT as switch · bipolar-vs-unipolar — each asked once; cheap 4-mark preparation.
- DC analysis + load line — never standalone, always embedded (they are the marks *inside* the design/numerical questions).
- Ebers-Moll — never asked yet; prepare a 4-mark version precisely because it's the untouched syllabus item.

6. Common Mistakes (master list)

- $\beta = \alpha/(1+\alpha)$ instead of $\alpha/(1-\alpha)$.
- Forgetting $(\beta+1)R_E$ in base-loop KVL (DC) and $(\beta+1)$ reflection (AC).
- Confusing BJT saturation with MOSFET saturation.
- Designing a CC stage with a collector resistor.
- Computing R_1 before R_2 (R_2 comes from the stiffness rule).

6. $V_B = V_E$ (forgetting +0.7).
7. Skipping DC analysis before small-signal work (no $I_E \Rightarrow$ no r_e).
8. Not drawing the demanded AC equivalent circuit.
9. Wrong A_v signs (CE $-$, CB $+$, CC $+$ and <1).
10. Using $I_C = \beta I_B$ inside saturation.
11. Ignoring the given r_o (fail to check $r_o \geq 10(R_C + R_E)$).
12. Using v_{be} -scale $\gg V_T$ while still applying small-signal formulas (state the validity condition).

7. Numerical Problem Types You Must Be Able to Solve

Type	Given $\rightarrow$ Find	Template
N1 Bias design	V_{CC} , I_C/I_E , $\beta \rightarrow$ all resistors	\$1.6-D recipe + Designs 1–4
N2 Bias analysis	Circuit resistors $\rightarrow I_{CQ}$, V_{CEQ}	\$1.6-C exact formula / \$1.3 procedure
N3 Amplifier numerical	Circuit + β (+ r_o) $\rightarrow Z_i$, Z_o , A_v	\$1.9-C worked PYQ
N4 Parameter calc	Q-point $\rightarrow g_m$, r_{π} , r_e	\$1.5
N5 Switch design	V_{CC} , R_C , $\beta \rightarrow$ required I_B / R_B	\$1.10
N6 $\alpha \leftrightarrow \beta$ / current juggling	any two of I_B , I_C , I_E , α , $\beta \rightarrow$ rest	\$1.1-D

Complete BJT PYQ Bank

(Every Chapter-1 question from the four provided papers; wording preserved; organized by year. Questions 3–10+ of each paper belong to Chapters 2–5 and are excluded.)

2083 Baishakh — Back (New Course), ENEX 151

Q	Question (verbatim)	Marks	Topic	Type
1	Why BJT is bipolar and FET is unipolar? Design β -independent type DC biased common collector amplifier. Given parameters are: $V_{CC} = 20$ V, $I_C = 2$ mA, and $\beta = 100$. Use firm biasing method.	[1+5]	Device physics + CC bias design	Theory + Design
2	Explain the role of bypass capacitor in amplifier circuits. Draw the small signal equivalent circuit of a common base amplifier circuit and find its input impedance, output impedance and voltage gain.	[2+5]	Bypass capacitor + CB small-signal	Theory + Derivation

2082 Bhadra — Regular (New Course), ENEX 151

Q	Question (verbatim)	Marks	Topic	Type
1	What is the significance of β -independency in a voltage divider biasing circuit? Design a β -independent voltage divider bias circuit for the given parameters: $V_{CC} = 15$ V, $I_E = 1$ mA and $\beta = 100$.	[1+5]	β -independence + bias design	Theory + Design
2	Explain the role of coupling capacitors in amplifier circuits. Draw the small signal equivalent circuit of a common emitter amplifier circuit and find its input impedance, output impedance and voltage gain.	[2+5]	Coupling capacitors + CE small-signal	Theory + Derivation

2082 Baishakh — Back (New Course), EX 151

Q	Question (verbatim)	Marks	Topic	Type
1	Design a β -independent voltage divider bias circuit using the appropriate guidelines for the given parameters: $V_{CC} = 12\text{ V}$, $I_E = 1\text{ mA}$, and $\beta = 100$.	[4]	Bias design	Design
2	Determine input resistance, voltage gain and output resistance of given common emitter BJT Amplifier. $\beta = 120$ and $r_o = 40\text{ k}\Omega$. (Circuit: $V_{CC} = +20\text{ V}$, $R_B = 470\text{ k}\Omega$, $R_C = 2.2\text{ k}\Omega$, $R_E = 0.56\text{ k}\Omega$, coupling C_1 , C_2 .)	[5]	CE-with- R_E small-signal	Numerical
3	Explain the operation principle of a BJT as a switch with the necessary diagrams.	[4]	Switch (cutoff/saturation)	Theory

2081 Ashwin — Regular (New Course - 2080 Batch), EX 151

Q	Question (verbatim)	Marks	Topic	Type
1	Design β independent type dc biased common emitter amplifier. Given parameters $V_{CC} = 20\text{ V}$, $I_C = 1.5\text{ mA}$ and $\beta = 110$. Use stiff biasing method.	[4]	CE bias design	Design
2	Why is a common collector BJT amplifier also known as an emitter follower? Draw the small signal model of common collector amplifier circuit and derive expressions for input resistance, voltage gain, and current gain.	[1+4]	Emitter follower + CC small-signal	Theory + Derivation
3	Derive trans-conductance of BJT.	[4]	g_m derivation	Derivation

PYQ → What to Study

PYQ	Topic Tested	Notes Section(s) to Study	Required Skill
2083 Q1(a) [1]	Bipolar vs unipolar conduction	§1.1-C (boxed answer)	State both carrier arguments
2083 Q1(b) [5]	CC bias design, firm rule	§1.6-C, §1.6-D, Design 3 ; swing logic §1.4	Design R_E , R_1 , R_2 with $V_E = V_{CC}/2$; no R_C
2083 Q2(a) [2]	Bypass capacitor	§1.8-C (+ numerical proof §1.9-C)	Explain DC-open/AC-short & gain restoration
2083 Q2(b) [5]	CB small-signal	§1.7-C (T model), §1.9-D ; table §1.9-F	Draw AC model; derive $Z_i \approx r_e$, $Z_o \approx R_C$, $A_v = +R_C/r_e$
2082 Bhadra Q1(a) [1]	Significance of β -independence	§1.6-C (boxed answer)	State variability of β + resistor-set I_E
2082 Bhadra Q1(b) [5]	Voltage-divider design	§1.6-D, Design 1 ; prerequisite §1.3	Full recipe with stated rules
2082 Bhadra Q2(a) [2]	Coupling capacitors	§1.8-B	DC-block/AC-couple + Q-point protection
2082 Bhadra Q2(b) [5]	CE small-signal	§1.7-B (π model), §1.9-B ; parameters §1.5	Draw model; derive Z_i , Z_o , A_v with signs
2082 Baishakh Q1 [4]	Voltage-divider design	§1.6-D, Design 4	Full recipe
2082 Baishakh Q2 [5]	CE-with- R_E numerical	§1.3-B (DC) → §1.9-C (full solution); r_o check §1.2-C	DC → r_e → reflection rule → numbers; $r_o \geq 10(R_C + R_E)$ check

PYQ	Topic Tested	Notes Section(s) to Study	Required Skill
2082 Baishakh Q3 [4]	BJT as switch	§1.10 ; regions §1.2-B; load line §1.4	Circuit + both states + $I_B > I_{C(sat)}/\beta$ + diagram
2081 Q1 [4]	CE bias design, stiff rule	§1.6-D, Design 2	Full recipe, quote stiff rule
2081 Q2(a) [1]	Why "emitter follower"	§1.9-E boxed answer	v_E follows v_B , $A_v \approx 1$
2081 Q2(b) [4]	CC derivations	§1.7-C, §1.9-E ; reflection rule §1.9-C	Derive R_{in} , A_v , A_i (and Z_o)
2081 Q3 [4]	Transconductance derivation	§1.5-B ; law from §1.1-D	4-step derivation with validity condition

Final Coverage Audit

Syllabus coverage — ✓ all eleven headings 1.1–1.11 present as the document's backbone, in syllabus order; nothing skipped (1.2 load-line and 1.11 Ebers-Moll included despite zero PYQs, per Rule 4).

Reference coverage — ✓ Sedra-Smith supplied the exponential-law/ g_m framework, π -T models, Early effect, Ebers-Moll, and the one-third design alternative; Boylestad-Nashelsky supplied $r_e \approx 26 \text{ mV}/I_E$, the exact $470\text{k}/2.2\text{k}/0.56\text{k}$ unbypassed- R_E analysis method and the $r_o \geq 10(R_C + R_E)$ check; Floyd supplied the stiff-divider rule ($R_2 \leq \beta R_E/10$) and switch treatment; Millman-Halkias/Bell corroborate bias-stability and configuration comparisons. ✓ Notation unified to I_B , I_C , I_E , V_{BE} , V_{CE} , α , β , g_m , r_{π} , r_e , r_o ; the 25 mV vs 26 mV V_T convention and $h_{FE} \equiv \beta$ equivalences flagged.

PYQ coverage — ✓ 10 BJT questions (15 sub-parts) identified across 4 papers; ✓ each appears inside its syllabus section; ✓ all appear in the Complete PYQ Bank; ✓ all mapped in PYQ → What to Study. Cross-topic questions handled: 2082 Baishakh Q2 spans DC analysis + small-signal (placed in §1.9, DC prerequisite cross-referenced to §1.3); 2083 Q1 spans device physics + biasing (split across §1.1/§1.6).

Exam preparation — ✓ derivations table (Dv1–Dv9) distinguishes *important derivations* from *results to remember*; ✓ formula boxes per section + consolidated sheet; ✓ 9 exam diagrams flagged; ✓ 6 numerical templates; ✓ **HOT** repeated topics identified (bias design 4/4, small-signal 4/4); ✓ priority labels ●●● assigned from PYQ evidence.

End of Chapter 1 notes.